LECTURE 7
INSTRUMENTAL VARIABLES ESTIMATION

(A) Endogeneities in Multiple Regression Model

(B) Inconsistency of OLS with Endogeneity

(C) Instrumental Variables Estimation in Simple Regression Model

(D) Instrumental Variables Estimation in Multiple Regression Model

(A) Endogeneities in Regression Model

- Consider a multiple regression model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_K x_{iK} + u_i, \qquad i = 1, 2, \ldots, n$$

Question: What are the four assumptions needed for unbiased OLS estimators?

- Suppose x_1 is correlated with u, i.e., $Cov(x_1, u) \neq 0$. x_1 is endogenous. The key assumption $E(u|x_1, \ldots, x_k) = E(u) = 0$, requiring that all factors in the unobserved error term be __uncorrelated__ with the explanatory variables is violated.

- The OLS estimator $\hat{\beta}_1$ (and very likely other estimator $\hat{\beta}_0$, $\hat{\beta}_2$, $\hat{\beta}_3$, …, and $\hat{\beta}_K$) are biased and __incosistent__. The OLS result is misleading.

- When $E(u|x_1, \ldots, x_k) = E(u) = 0$ holds, $x_1, \ldots, x_k$ are **exogenous** explanatory variables. If x_j is correlated with u for any reason, then x_j is an **endogenous** explanatory variable.

Example (Return to Education): (dataset:WAGE2.DTA)

Population Model: $\ln(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + u$

Estimated Model: $\ln\widehat{(wage)} = 5.50 + 077educ + 0.019exper$

Clearly, *educ* is endogenous because $Cov(educ, u) \neq 0$. The OLS estimator for $educ$ is biased and inconsistent; the estimated coefficient for $educ$ is misleading.

- Endogeneity arises when one explanatory is correlated with the error term u:

$$Cov(x_1, u) \neq 0$$

- Possible sources of endogeneity:
  - Omitted variables bias
  - Measurement error
  - Selection bias
  - Misspecification of the function form
  - Reverse causality (or simultaneity bias)

Remarks

Omitted variable bias:

We cannot control for everything in model, e.g., ability, omitted in the regression. In the example of return to education, if the omitted unobserved variable (e.g., ability) is positively correlated with wage and schooling (high-ability people tend to go to college and earn more), then the estimated return to education will be biased *upward*. That is, we overestimate the effect of schooling, because of the omitted variable (ability).

Sample selection bias

- The sample used in the analysis is *nonrandom*. Some groups of population are *more* likely to be in the sample than others.
- For example, less healthy people are more likely to go to hospital than healthy people.
- When we look at the effect of hospitialization on health, the survey data from hospitals do not reflect the whole picture of the population.
- With nonrandom sample, the error term u cannot be regarded as pure random shocks, implying that the resulting OLS could be biased and inconsistent.

- In the Return to Education example, $educ$ is endogenous variable, $Cov(educ, u) \neq 0$. There is an omitted variable bias on $\hat{\beta}_1$,

- Suppose one of the explanatory variables x_1 is correlated with an omitted variable (OV), e.g., $ability$, where the OV is absorbed by the error u, and $u = \beta_3 OV + \varepsilon$.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + (\beta_3 OV + \varepsilon)$$

Clearly, $Cov(x_1, u) \neq 0$ because

$$Cov(x_1, \beta_3 OV + \varepsilon) = \beta_3 Cov(x_1, OV) + Cov(x_1, \varepsilon) \neq 0$$

- How do we remove (or control) the OV? In other words, how do we fix the endogeneity problem?
  (a) Proxy Variable (We can use IQ as a substitute for *ability*)
  (b) Instrumental Variable
  (c) Panel Data

(B) Inconsistency of OLS with Endogeneity

- Let's consider a simple regression model that suffers endogeneity problem first.

$$y_i = \beta_0 + \beta_1 x_i + u_i,\ \ i = 1,2,..n, \text{ where } Cov(x,u) \neq 0.$$

- The OLS estimators of β_0 and β_1 are $\hat{\beta}_0$ and $\hat{\beta}_1$. These two OLS estimators minimize the sum of squared residuals

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{\widehat{Cov(x,y)}}{\widehat{Var(x)}}$$

where $\hat{\beta}_1$ is the sample covariance between x and y divided by the sample variance of x. (Refer to Assignment 2 Question 1 for the $\hat{\beta}_1$ derivation.)

- By the law of large number, when the sample size n is large,

  $\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \widehat{Cov(x,y)} \approx Cov(x,y)$, and

  $\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \widehat{Var(x)} \approx Var(x)$.

Therefore, $\hat{\beta}_1 = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2} \approx \frac{Cov(x,y)}{Var(x)}$

- Since $Cov(x,y) = Cov(x, \beta_0 + \beta_1 x + u) = \beta_1 Var(x) + Cov(x,u)$,

$$\hat{\beta}_1 \approx \frac{Cov(x,y)}{Var(x)} = \beta_1 + \frac{Cov(x,u)}{Var(x)} \neq \underline{\text{Beta 1}}$$

- $\hat{\beta}_1$ could be very far from the true value β_1 even if $n \to \infty$. Endogeneity makes OLS estimators unreliable.

(C) Instrumental Variables (IV) Estimation in Simple Regression Model

- When the proxy variable is not available, for example IQ is not available for the omitted variable (OV) -- ability, we could consider to use instrumental variable(s) to remove the OV effect through the endogenous variable x on the dependent variable y.

- Consider a simple regression model $y_i = \beta_0 + \beta_1 x_i + u_i,\ i = 1,2,..n$, where $Cov(x, u) \neq 0$. We might fix the endogeneity problem if we can find a new variable, say z, which is exogenous. In other words, $Cov(z, u) = \underline{\ 0\ }$

- The covariance relationship between z and y is

$$Cov(z, y) = Cov(z, \beta_0 + \beta_1 x + u) = \beta_1 Cov(z, x) + Cov(z, u) = \beta_1 Cov(z, x).$$

If we solve for β_1,

$$\beta_1 = \frac{Cov(z, y)}{Cov(z, x)}$$

- To estimate the unknown β_1, we use sample counterparts of $Cov(z, y)$and $Cov(z, x)$, i.e.,

$$\hat{\beta}_{1,IV} = \frac{\widehat{Cov(z,y)}}{\widehat{Cov(z,x)}} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})(x_i - \bar{x})} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})y_i}{\sum_{i=1}^{n}(z_i - \bar{z})x_i} \text{ and } \hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x}$$

- $\hat{\beta}_{1,IV}$ and $\hat{\beta}_{0,IV}$ are the instrumental estimators for β_1 and β_0.

- The bottom line is the instrumental variable z should satisfy two conditions:
  (1) $Cov(z, u) = 0$ (Instrument Exogeneity)
  (2) $Cov(z, x) \neq 0$ (Instrument Relevance)
  [Recall: $\ln(wage) = \beta_0 + \beta_1 educ + u$]

- The instrumental variable (and the proxy variable) can both be used to solve (or at least mitigate) the endogeneity problem.

Empirical example: Return to Education (dataset: WAGE2.DTA)

$$\ln\,(wage) = \beta_0 + \ \beta_1 educ + u$$

Stata > regress lwage educ

```
. regress lwage educ

      Source |       SS       df       MS              Number of obs =     935
-------------+------------------------------           F(  1,   933) =  100.70
       Model |  16.1377042     1  16.1377042           Prob > F      =  0.0000
    Residual |  149.518579   933  .160255712           R-squared     =  0.0974
-------------+------------------------------           Adj R-squared =  0.0964
       Total |  165.656283   934  .177362188           Root MSE      =  .40032

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0598392   .0059631    10.03   0.000     .0481366    .0715418
       _cons |   5.973063   .0813737    73.40   0.000     5.813366    6.132759
------------------------------------------------------------------------------
```

(a) *ability* is omitted and absorbed into the error term u. Since *ability* is correlated with $educ$, $educ$ is endogenous. The OLS result is not reliable.

(b) We could use the IV approach if an exogenous variable z is available. A good instrument variable should satisfy two conditions, namely instrument exogeneity and instrument relevance.

(c) A possible candidate is *number of siblings*. Why?

Stata> ivreg lwage (educ= sibs)

```
Instrumental variables (2SLS) regression

      Source |       SS       df       MS              Number of obs =     935
-------------+------------------------------           F(  1,   933) =   21.59
       Model | -1.51973315     1 -1.51973315           Prob > F      =  0.0000
    Residual |  167.176016   933  .179181154           R-squared     =       .
-------------+------------------------------           Adj R-squared =       .
       Total |  165.656283   934  .177362188           Root MSE      =   .4233

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1224326   .0263506     4.65   0.000     .0707194    .1741459
       _cons |   5.130026   .3551712    14.44   0.000     4.432999    5.827053
------------------------------------------------------------------------------
Instrumented:  educ
Instruments:   sibs
------------------------------------------------------------------------------
```

<u>Benefit</u> of the IV estimator: consistent (It's reliable when *n* is large)

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^{n}(z_i-\bar{z})y_i}{\sum_{i=1}^{n}(z_i-\bar{z})x_i} = \frac{\sum_{i=1}^{n}(z_i-\bar{z})(\beta_0+\beta_1 x_i+u_i)}{\sum_{i=1}^{n}(z_i-\bar{z})x_i} = \beta_1 + \frac{\frac{1}{n-1}\sum_{i=1}^{n}(z_i-\bar{z})u_i}{\frac{1}{n-1}\sum_{i=1}^{n}(z_i-\bar{z})x_i}$$

When *n* is large,

$$\frac{1}{n-1}\sum_{i=1}^{n}(z_i-\bar{z})u_i \approx Cov(z,u) = 0$$

$$\frac{1}{n-1}\sum_{i=1}^{n}(z_i-\bar{z})x_i \approx Cov(z,x)$$

so, $\hat{\beta}_{1,IV} \approx \beta_1 + 0 = \beta_1$

<u>Cost</u> of the IV estimator: <u>large</u> variance

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})y_i}{\sum_{i=1}^{n}(z_i - \bar{z})x_i} = \beta_1 + \frac{\sum_{i=1}^{n}(z_i - \bar{z})u_i}{\sum_{i=1}^{n}(z_i - \bar{z})x_i}$$

So its approximate (by large sample) variance:

$$Var(\hat{\beta}_{1,IV}) \approx \frac{\sigma^2 \sum_{i=1}^{n}(z_i - \bar{z})^2}{\left(\sum_{i=1}^{n}(z_i - \bar{z})x_i\right)^2}$$

$$= \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \times \frac{\left(\sum_{i=1}^{n}(z_i - \bar{z})^2\right)\left(\sum_{i=1}^{n}(x_i - \bar{x})^2\right)}{\left(\sum_{i=1}^{n}(z_i - \bar{z})x_i\right)^2}$$

$$= Var(\hat{\beta}_{1,OLS}) \times \frac{1}{\hat{\rho}_{z,x}^2} \geq Var(\hat{\beta}_{1,OLS})$$

where $\hat{\rho}_{z,x}^2 = \frac{\left(\frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})x_i\right)^2}{\left(\frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})^2\right)\left(\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2\right)} \leq 1$ is the (squared) correlation coefficient between z and x.

Remarks:

(a) When the instrumental variable z is a good representative for x, $\hat{\rho}^2_{z,x}$ is very big.

(b) When x is exogenous, it can be regarded as an IV for itself, i.e., $z = x$.

In this case, $\hat{\rho}^2_{z,x} = 1$. Clearly, for this case IV = OLS, because

$$\hat{\beta}_{1,IV} = \frac{\sum_{i=1}^{n}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{n}(z_i - \bar{z})(x_i - \bar{x})} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})} = \hat{\beta}_{1,OLS}, \text{ and}$$

$$Var(\hat{\beta}_{1,IV}) \approx= Var(\hat{\beta}_{1,OLS}) \times \frac{1}{\hat{\rho}^2_{z,x}} = Var(\hat{\beta}_{1,OLS}).$$

(c) In order the instrumental variables estimation to work well in practice, we need (1) instrument exogeneity, (2) instrument relevance, and (3) ***large*** sample size.

- A practical and important question:
  How do I know my instrumental variable z is good? (How do I know the instrument relevance and exogeneity conditions are met?)

- To check the instrument relevance, we regress the endogenous variable x on the instrumental variable z.

$$x = \pi_0 + \pi_1 z + v, \quad \text{where } v \text{ is an error term}$$

After the estimation, we can use the usual t statistic to test for

$$H_0: \pi_1 = 0 \text{ against } H_1: \pi_1 \neq 0.$$

[Recall: Calculate $t = \frac{\hat{\pi}_1 - 0}{s.e(\hat{\pi}_1)}$ and compare the answer to the critical values.]

- If we reject the null hypothesis, then $Cov(z, x) \neq 0$. The instrument relevance condition is met.

Consider the Return to Education example. We run the following regression to check the instrument relevance condition.

$$educ = \pi_0 + \pi_1 sibs + v$$

```
. reg educ sibs

      Source |       SS           df       MS      Number of obs   =       935
-------------+----------------------------------   F(1, 933)       =     56.67
       Model |  258.055048         1  258.055048   Prob > F        =    0.0000
    Residual |   4248.7642       933  4.55387374   R-squared       =    0.0573
-------------+----------------------------------   Adj R-squared   =    0.0562
       Total |  4506.81925       934  4.82528828   Root MSE        =     2.134

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        sibs |  -.2279164   .0302768    -7.53   0.000     -.287335   -.1684979
       _cons |   14.13879   .1131382   124.97   0.000     13.91676    14.36083
------------------------------------------------------------------------------
```

(a) $t = \frac{\hat{\pi}_1 - 0}{s.e(\hat{\pi}_1)} = \frac{-.228-0}{0.03} = -7.53$

(b) The statistic value is much smaller than -1.96 (and the P-value 0.000 is smaller than the 5% significance level). The sibs is statistically significant.

(c) So, $Cov(z, x) = Cov(sibs, educ) \neq 0$

(D) Instrumental Variables (IV) Estimation in Multiple Regression Model

- Consider a more general case, for example one endogenous explanatory variable in a multiple regression model case.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u,$$

where y: dependent variable, x_1 is *endogenous*, x_2 is *exogenous*

Example: Return to Education

$$\ln(\text{wage}) = \beta_0 + \beta_1 educ + \beta_2 married + u$$

y: ln(wage)

x_1: *educ* (*endogenous* variable)

x_2: married(*exogenous* variable)

Need a **good** z_1 variable, which is an (outside) instrument for endogenous x_1; z_1 must be different from x_2.

The *instrument exogeneity* means

$$Cov(z_1, u) = 0$$

implying that

$$Cov(z_1, y) = \beta_1 Cov(z_1, x_1) + \beta_2 Cov(z_1, x_2) \qquad (1*)$$

Besides, $Cov(x_2, u) = 0$ implies that

$$Cov(x_2, y) = \beta_1 Cov(x_2, x_1) + \beta_2 Var(x_2) \qquad (2*)$$

Like in the simple regression model, replacing the covariance's and variance in (1*) and (2*) with their ___sample___ counterparts gives the IV estimators of β_1 and β_2, denoted by $\hat{\beta}_{1,IV}$, $\hat{\beta}_{2,IV}$. Similarly,

$$\hat{\beta}_{0,IV} = \bar{y} - \hat{\beta}_{1,IV}\bar{x}_1 - \hat{\beta}_{2,IV}\bar{x}_2$$

where "-" means sample average.

- One instrument is needed for one endogenous explanatory variable. This is called *exactly identified* case.

Stata> ivreg lwage (educ = sibs) married

```
. ivreg lwage (educ = sibs) married

Instrumental variables (2SLS) regression

      Source |       SS       df       MS              Number of obs =     935
-------------+------------------------------           F(  2,   932) =   21.89
       Model |  4.64837902     2  2.32418951           Prob > F      =  0.0000
    Residual |  161.007904   932  .172755262           R-squared     =  0.0281
-------------+------------------------------           Adj R-squared =  0.0260
       Total |  165.656283   934  .177362188           Root MSE      =  .41564

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1217837   .0258461     4.71   0.000     .0710603     .172507
     married |     .25493   .0452778     5.63   0.000     .1660718    .3437882
       _cons |   4.911102   .3601172    13.64   0.000     4.204368    5.617837
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Instrumented:  educ
Instruments:   married sibs
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- To check the instrument relevance, we regress the endogenous variable x_1 on z and all other exogenous variable(s) which is only x_2 here.

$$x = \pi_0 + \pi_1 z + \pi_2 x_2 + v, \quad \text{where } v \text{ is an error term}$$

After the usual OLS estimation, we can use the usual t statistic to test for

$$H_0: \pi_1 = 0 \text{ against } H_1: \pi_1 \neq 0.$$

- Why do we need to include x_2 when we test the instrumental relevance condition for z_1?

- z_1 could be correlated with x_2. An extreme case: z_1 and x_2 are perfectly correlated, say $z_1 = 2x_2$. Intuitively, having z_1 doesn't bring additional information in this case.

  Instrument relevance requires that after partialling out the effect of x_2, x_1 and z_1 are still correlated.

Intuition:

We decompose endogenous x_1 into two parts:

$$x_1 = \hat{x}_1 + v,\ Cov(\hat{x}_1, u) = 0 \text{ and } Cov(v, u) \neq 0$$

where $\hat{x}_1$ is predicted by the instrumental variable (z_1) and exogenous regressors (x_2), i.e., $\hat{x}_1 = \hat{\pi}_0 + \hat{\pi}_1 z + \hat{\pi}_2 x_2$.